

Printable Formula Sheets

The counts, coverage figures and rules testers must be able to work out.

Using This Sheet

The Foundation exam includes a small number of calculation-style questions. Learn these rules, then practise the worked micro-examples until you can reproduce them without hesitation. Print this sheet and keep it beside you while revising.

Boundary Value Analysis (BVA)

For a valid range, the two-value BVA uses the minimum and maximum boundary values. The three-value BVA also adds the value just outside each boundary. Test the boundaries because defects cluster at the edges.

Range	Two-value boundaries	Three-value boundaries
1 to 100	1 and 100	0, 1, 2 ... 99, 100, 101
18 to 65 (age)	18 and 65	17, 18, 19 ... 64, 65, 66
0.00 to 9.99	0.00 and 9.99	-0.01, 0.00, 0.01 ... 9.98, 9.99, 10.00

Equivalence Partitions per Range

A single bounded range typically yields three partitions: one valid partition (inside the range) and two invalid partitions (below the minimum and above the maximum). Test one representative value from each.

Range	Valid partition	Invalid partitions
1 to 100	e.g. 50	below (e.g. 0) and above (e.g. 101)
18 to 65	e.g. 30	below (e.g. 17) and above (e.g. 66)

Decision Table — Rule Count

Rule count = 2 raised to the number of conditions (for binary Yes/No conditions). Each rule is one column combining a state of every condition.

Conditions (binary)	Formula 2^n	Number of rules (columns)
2 conditions	2^2	4 rules
3 conditions	2^3	8 rules
4 conditions	2^4	16 rules
5 conditions	2^5	32 rules

Coverage — Statement and Branch

Coverage = (items exercised / total items) x 100%.

Metric	Formula	Worked example
Statement coverage	(statements executed / total statements) x 100	7 of 10 statements run = 70%
Branch coverage	(branches executed / total branches) x 100	3 of 4 branches run = 75%
Test coverage (any item)	(items tested / total items) x 100	18 of 20 requirements tested = 90%

Defect Density

Defect density = number of defects / size of the component or system. Size is commonly expressed in thousands of lines of code (KLOC) or in function points.

Defects	Size	Defect density
40 defects	20 KLOC	2 defects per KLOC
15 defects	30 function points	0.5 defects per function point

Pass Rate and Failure Rate

Pass rate = (tests passed / tests executed) x 100%. Failure rate = (tests failed / tests executed) x 100%. The two add up to 100% of executed tests.

Tests executed	Passed	Failed	Pass rate
200	180	20	90%
50	45	5	90%
80	60	20	75%

Defect Detection Percentage (DDP)

DDP = (defects found by testing / (defects found by testing + defects found later)) x 100%. It measures how effective a test phase was at finding defects.

Found in testing	Found after release	DDP
90	10	90%

Found in testing	Found after release	DDP
80	20	80%

COMMON MISTAKE

Coverage is not the same as quality. 100% statement coverage only proves every line ran — it does not prove every combination or requirement was tested. High coverage with weak assertions still misses defects.

INSIDE STLC PRO TIP

For decision tables, always start from 2^n columns, then collapse impossible or duplicate combinations. Working out the full count first stops you from silently missing a rule.